Group Topic: **Big Data Analysis for Rainfall-Based Flood Risk Prediction in Albay using SYNOP Data**	Week 2 Task 2
Git Repo/ Colab Link: https://github.com/PotatoKevz/About-bda	Date: April 24, 2026

Objectives

- Conceptualize the data gathering method
- Look for online sites that can be requested for SYNOP data
- Decode the downloaded raw data reports in a CSV format
- Clean the decoded data reports
- Visualize the cleaned data reports

Tasks Completed

1. **Looked for sites that can be requested by SYNOP Dataset, and the group arrived at OGIMET (*Organización Meteorológica Iberoamericana)*, an online platform that provides access to historical weather observations, especially SYNOP reports from meteorological stations worldwide.**

2. **Scraped data from weather stations near Albay; Albay (Lezapi City), Camarines Norte (Daet), Camarines Sur (Naga), Catanduanes (Virac), Masbate, Romblon, and Quezon (Infanta and Tayabas) Stations:**

98427 Tayabas (Quezon) 75k.csv	jab2023-6181-30913	Apr 23	8.2 MB	⋮
98434 Infanta (Quezon) 75k.csv	jab2023-6181-30913	Apr 23	8.2 MB	⋮
98440 Camarines Norte 76k.csv	jab2023-6181-30913	Apr 23	8.5 MB	⋮
98442 Camarines Sur Station 11k.csv	jab2023-6181-30913	Apr 23	3.5 MB	⋮
98444 Albay Station 76k.csv	jab2023-6181-30913	Apr 23	8.4 MB	⋮
98446 Catanduanes 76k.csv	jab2023-6181-30913	Apr 23	8.2 MB	⋮
98536 Romblon 74k.csv	jab2023-6181-30913	Apr 23	7.9 MB	⋮
98543 Masbate 76k.csv	jab2023-6181-30913	Apr 23	8 MB	⋮

Scraped Datasets from OGIMET

	A	B	C	D	E	F	G
1	WMOIND	YEAR	MONTH	DAY	HOUR	MIN	REPORT
2	98444	2000	1	1	0	0	AAXX 01001 98444
3	98444	2000	1	1	3	0	AAXX 01031 98444
4	98444	2000	1	1	6	0	AAXX 01061 98444
5	98444	2000	1	1	9	0	AAXX 01091 98444
6	98444	2000	1	1	12	0	AAXX 01121 98444
7	98444	2000	1	1	15	0	AAXX 01151 98444
8	98444	2000	1	1	18	0	AAXX 01181 98444
9	98444	2000	1	2	0	0	AAXX 02001 98444
10	98444	2000	1	2	3	0	AAXX 02031 98444
11	98444	2000	1	2	6	0	AAXX 02061 98444
12	98444	2000	1	2	9	0	AAXX 02091 98444
13	98444	2000	1	2	12	0	AAXX 02121 98444
14	98444	2000	1	2	15	0	AAXX 02151 98444
15	98444	2000	1	2	18	0	AAXX 02181 98444
16	98444	2000	1	2	21	0	AAXX 02211 98444
17	98444	2000	1	3	0	0	AAXX 03001 98444
18	98444	2000	1	3	3	0	AAXX 03031 98444
19	98444	2000	1	3	6	0	AAXX 03061 98444
20	98444	2000	1	3	9	0	AAXX 03091 98444
21	98444	2000	1	3	12	0	AAXX 03121 98444
22	98444	2000	1	4	0	0	AAXX 04001 98444
23	98444	2000	1	4	3	0	AAXX 04031 98444
24	98444	2000	1	4	6	0	AAXX 04061 98444
25	98444	2000	1	4	9	0	AAXX 04091 98444
26	98444	2000	1	4	12	0	AAXX 04121 98444
27	98444	2000	1	4	15	0	AAXX 04151 98444

CSV Content of Albay Station (Raw SYNOT Data Report)

```python
C: > Users > romeb > Desktop > Big Data Analysis > Programs > flood-prediction > ogimet_synop_scraper.py > _chunks
BASE_URL = "http://www.ogimet.com/cgi-bin/getsynop"
COLUMNS  = ["WMOIND", "YEAR", "MONTH", "DAY", "HOUR", "MIN", "REPORT"]

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(levelname)s] %(message)s",
    datefmt="%Y-%m-%d %H:%M:%S",
)
log = logging.getLogger(__name__)

# Mimic a real browser — OGIMET rejects bot-like user agents
HTTP_HEADERS = {
    "User-Agent": (
        "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
        "AppleWebKit/537.36 (KHTML, like Gecko) "
        "Chrome/124.0.0.0 Safari/537.36"
    ),
    "Accept"          : "text/plain, text/html, */*",
    "Accept-Language" : "en-US,en;q=0.9",
    "Connection"      : "keep-alive",
}


# ── helpers ──────────────────────────────────────────────────────────

def _fmt(dt: datetime) -> str:
    return dt.strftime("%Y%m%d%H%M")


def _chunks(start_year: int, end_year: int, days: int) -> Iterator[Tuple[datetime, datetime]]:
    """Yield (begin, end) datetime pairs in `days`-sized steps."""
    now   = datetime.now(timezone.utc).replace(tzinfo=None)
```

Code Snippet of Data Scraper

3. Decoded the raw SYNOP data reports:

DecodedAlbay.csv	me	5:21 PM me	5 MB
DecodedCamNorte.csv	me	5:21 PM me	4.2 MB
DecodedCamSur.csv	me	5:21 PM me	2.4 MB
DecodedCatanduanes.csv	me	5:21 PM me	4.2 MB
DecodedInfanta.csv	me	5:21 PM me	4.1 MB
DecodedMasbate.csv	me	5:21 PM me	4.1 MB
DecodedRomblon.csv	me	5:21 PM me	3.9 MB
DecodedTayabas.csv	me	5:22 PM me	4.1 MB

Decoded Datasets

	A	B	C	D	E	F	G	H	I	J	K	L
1	station_id	datetime	lat	lon	temp	pressure	humidity	wind_speed	wind_dir	cloud_cover	visibility_m	rain_3h
2	98444	2000-01-01 0:00	13.15	123.73	24.8	1011.7	98.8	3	40	8	15000	51
3	98444	2000-01-01 3:00	13.15	123.73	26.2	1011.1	92.6	2	40	8	17000	
4	98444	2000-01-01 6:00	13.15	123.73	28	1008.5	86.8	5	50	6	20000	9
5	98444	2000-01-01 9:00	13.15	123.73	28.2	1008.7	84.3	4	90	6	20000	
6	98444	2000-01-01 12:00	13.15	123.73	27.1	1011.2	86.2	4	40	6	20000	
7	98444	2000-01-01 15:00	13.15	123.73	25.4	1011.5	93.6	2	40	6	20000	
8	98444	2000-01-01 18:00	13.15	123.73	26.2	1009.4	82.1	3	40	5	20000	
9	98444	2000-01-02 0:00	13.15	123.73	25.7	1011.4		2	70	7	20000	10
10	98444	2000-01-02 3:00	13.15	123.73	29.2	1011.7	75.4	2	40	7	23000	
11	98444	2000-01-02 6:00	13.15	123.73	25.5	1009.6	95.3	1	90	7	15000	28
12	98444	2000-01-02 9:00	13.15	123.73	26.8	1009	88.3	2	40	7	20000	
13	98444	2000-01-02 12:00	13.15	123.73	25.6	1011	90.9	2	70	5	20000	
14	98444	2000-01-02 15:00	13.15	123.73	25	1011.7	90.8	2	40	4	20000	
15	98444	2000-01-02 18:00	13.15	123.73	25.2	1010	90.9	2	50	6	20000	
16	98444	2000-01-02 21:00	13.15	123.73	24	1009.7	97	3	40	8	17000	
17	98444	2000-01-03 0:00	13.15	123.73	24.8	1011.8	93.1	4	40	8	23000	38
18	98444	2000-01-03 3:00	13.15	123.73	25.8	1012	91.4	3	40	7	20000	
19	98444	2000-01-03 6:00	13.15	123.73	27	1009.7	85.2	3	40	7	23000	4
20	98444	2000-01-03 9:00	13.15	123.73	26.4	1009.6	89.3	4	40	5	20000	
21	98444	2000-01-03 12:00	13.15	123.73	25	1011.4	89.2	3	40	6	20000	0.3
22	98444	2000-01-04 0:00	13.15	123.73	26	1012.4	83.6	2	40	5	23000	4
23	98444	2000-01-04 3:00	13.15	123.73	28.9	1012.7	74	5	40	5	23000	
24	98444	2000-01-04 6:00	13.15	123.73	29.3	1010.1	73.6	3	20	6	23000	0

CSV Content of Albay Station (Decoded)

synop_decoder(1).py > parse_synop_v2

```python
# ----------------------------------------------------------------------

def decode_irix(group: str) -> dict:
    """Section 0 indicator group  iRiXhVV  (5 chars)."""
    out = {}
    if len(group) != 5:
        return out
    # iR – precipitation indicator  (ignored for output)
    # iX – station type / present-weather indicator  (ignored)
    # h  – height of cloud base (code table 1600)
    h_code = group[2]
    h_table = {"0": 0, "1": 50, "2": 100, "3": 200, "4": 300,
               "5": 600, "6": 1000, "7": 1500, "8": 2000, "9": 2500, "/": None}
    out["cloud_base_m"] = h_table.get(h_code)
    # VV – visibility (code table 4377)
    vv = group[3:5]
    out["visibility_m"] = _decode_vv(vv)
    return out


def _decode_vv(vv: str) -> float | None:
    """Convert VV code (00-99) to metres."""
    if "/" in vv:
        return None
    v = int(vv)
    if 0 <= v <= 50:
        return v * 100           # 00-50 → 0 … 5 000 m
    elif 51 <= v <= 80:
```

PROBLEMS DEBUG CONSOLE TERMINAL PORTS powershell

```
PS C:\Users\detectivejordan\OneDrive\Documents\Big Data Analysis Lessons\synop ogimet> python "synop_decoder(1).py" "9
```

Code Snippet of Decoder

4. Cleaned the decoded SYNOP datasets:

CleanedAlbay.csv	jab2023-6181-30913	6:37 PM	5.4 MB	⋮
CleanedCamNorte.csv	jab2023-6181-30913	6:42 PM	5.4 MB	⋮
CleanedCatanduanes.csv	jab2023-6181-30913	6:42 PM	5.3 MB	⋮
CleanedInfanta.csv	jab2023-6181-30913	6:43 PM	5.3 MB	⋮
CleanedMasbate.csv	jab2023-6181-30913	6:43 PM	5.4 MB	⋮
CleanedRomblon.csv	jab2023-6181-30913	6:43 PM	5.2 MB	⋮
CleanedTayabas.csv	jab2023-6181-30913	6:44 PM	5.3 MB	⋮

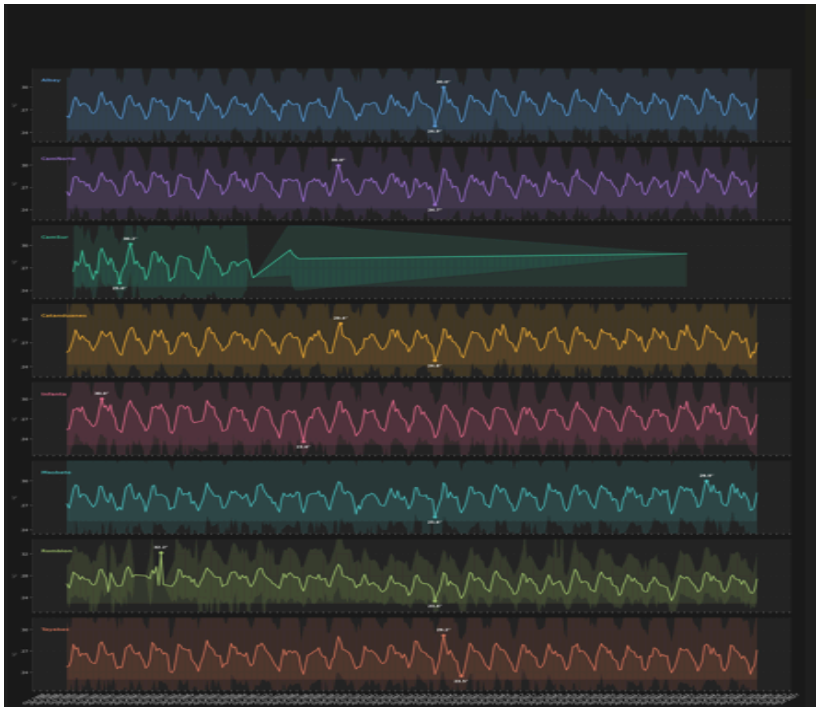
Cleaned Datasets

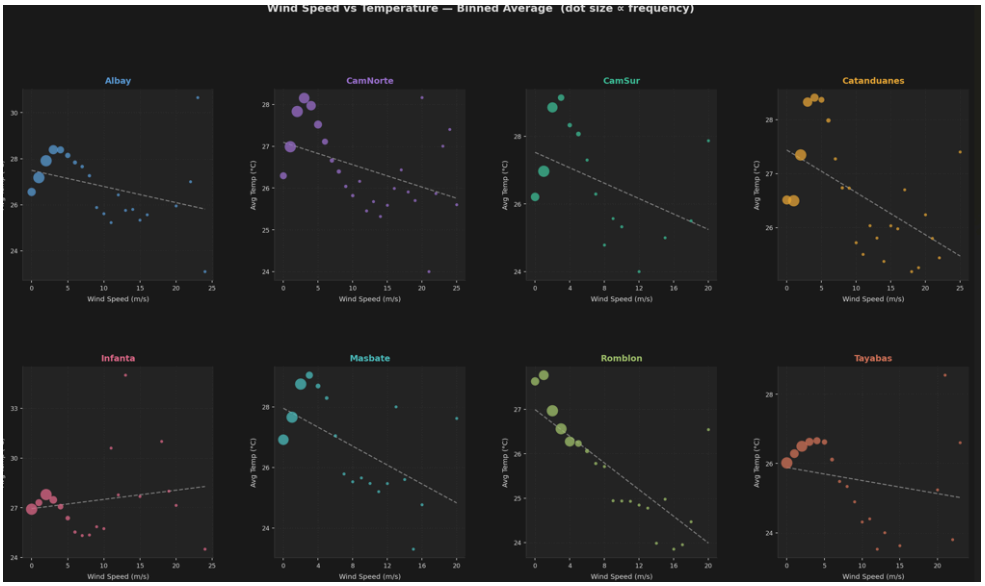
C: > Users > romeb > Desktop > Big Data Analysis > Programs > flood-prediction > synop_decoder.py > ...

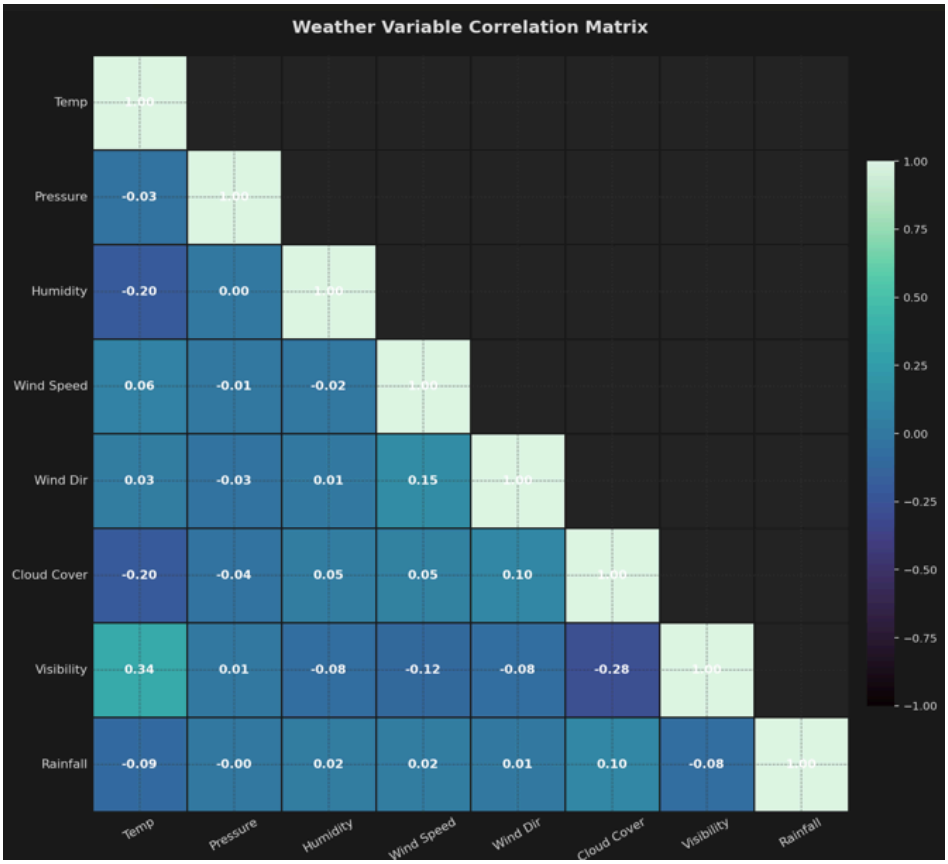
```python


def decode_irix(group: str) -> dict:
    """Section 0 indicator group  iRiXhVV  (5 chars).

    iR (precipitation indicator):
        0/1/2 ─ group 6 present somewhere  → no sentinel (use decoded value)
        3     ─ precip amount = 0, no group 6  → rain_sentinel = 0.0  (Case B)
        4     ─ station does not measure precip → rain_sentinel = None (Case A)
    """
    out = {}
    if len(group) != 5:
        return out
    ir = group[0]
    if ir == "3":
        out["rain_sentinel"] = 0.0   # "no rain reported" → 0.0 mm
    elif ir == "4":
        out["rain_sentinel"] = None  # truly missing (station doesn't measure)
    # iX - station type / present-weather indicator  (ignored)
    # h  - height of cloud base (code table 1600)
    h_code = group[2]
    h_table = {"0": 0, "1": 50, "2": 100, "3": 200, "4": 300,
               "5": 600, "6": 1000, "7": 1500, "8": 2000, "9": 2500, "/": None}
    out["cloud_base_m"] = h_table.get(h_code)
    # VV - visibility (code table 4377)
    vv = group[3:5]
    out["visibility_m"] = _decode_vv(vv)
    return out


def _decode_vv(vv: str) -> float | None:
    """Convert VV code (00-99) to metres (WMO Code Table 4377)."""
```

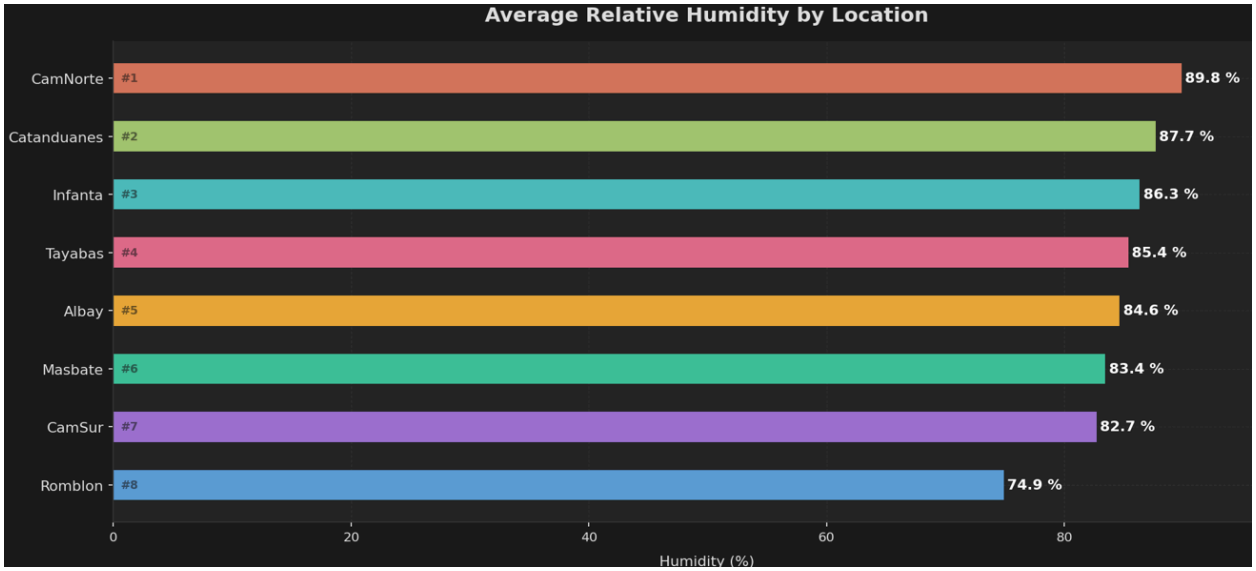
Code Snippet

5. Visualization of the datasets:






Weather Variable Correlation Matrix
Temp
Pressure
Humidity
Wind Speed
Wind Dir
Cloud Cover
Visibility
Rainfall
1.00
-0.03
-0.20 0.00
0.06 -0.01 -0.02
0.03 -0.03 0.01 0.15
-0.20 -0.04 0.05 0.05 0.10
0.34 0.01 -0.08 -0.12 -0.08 -0.28
-0.09 -0.00 0.02 0.02 0.01 0.10 -0.08
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Average Relative Humidity by Location
CamNorte #1 89.8 %
Catanduanes #2 87.7 %
Infanta #3 86.3 %
Tayabas #4 85.4 %
Albay #5 84.6 %
Masbate #6 83.4 %
CamSur #7 82.7 %
Romblon #8 74.9 %
0
20
40
60
80
Humidity (%)

Results and Analysis

The data collection phase yielded comprehensive historical records for the Bicol region. Initial analysis reveals:

- Consistency: Most stations show stable seasonal temperature fluctuations, though some gaps were noted in the Camarines Sur dataset.
- Correlation: A notable positive correlation (0.34) exists between temperature and visibility, while cloud cover shows a negative correlation (-0.20) with temperature.
- Humidity: The region maintains high moisture levels, with most stations averaging above 80% relative humidity, which is a critical factor for flood risk modeling.

Challenges and Solutions

Challenge: Difficulty in finding and gathering a large enough dataset (e.g., 500k+ records) for big data analysis.
Solution: Developed a **Python-based data scraper** to automatically collect historical records from the **OGIMET platform** , allowing the group to build a substantial database of weather observations from multiple stations.

Challenge: The raw data from weather stations is in a coded, "compressed" format (SYNOP) that is not readable by standard analysis tools.
Solution: Created a **custom decoder script** to translate these alphanumeric codes into a clean, structured CSV format with actual measurements for temperature, rainfall, and humidity.

Conclusion

This week's tasks successfully moved the project from the planning stage to active data management. By establishing an automated pipeline to scrape and decode SYNOP data, the group has secured the necessary high-fidelity information required for rainfall-based flood prediction. The objectives for data gathering, decoding, and initial visualization were fully met.